

Bitcoin Mining Pool Decision Guide

Introduction

Bitcoin mining has evolved from a simple process that could be performed on personal computers to an industrial-scale operation requiring specialized hardware. Mining pools allow miners to combine their computational resources and share the rewards, providing more consistent returns compared to solo mining. This guide will help you understand the different types of mining pools, their advantages, disadvantages, and important considerations.

Types of Mining Pools

1. Pay-Per-Share (PPS)

Description: Miners receive a fixed payment for each valid share they submit, regardless of whether the pool finds a block.

Advantages:

- Predictable, stable income
- Immediate payouts upon share submission
- No variance in payments
- Pool operator assumes all risk

Disadvantages:

- Usually lower overall returns compared to other payment models
- Higher pool fees (typically 2-4%)
- No additional rewards from transaction fees

Risk Level: Low

2. Full Pay-Per-Share (FPPS)

Description: Similar to PPS, but includes transaction fees in the payout calculation.

Advantages:

- All benefits of PPS
- Additional income from transaction fees
- More predictable than PPLNS during times of high transaction fees

Disadvantages:

- Still has higher pool fees than proportional systems
- May have slightly lower returns compared to PPLNS during stable mining periods

Risk Level: Low

3. Pay-Per-Last-N-Shares (PPLNS)

Description: Rewards are distributed based on the number of shares miners contributed in a specific time window before a block is found.

Advantages:

- Potentially higher returns than PPS/FPPS
- Lower pool fees (typically 1-2%)
- Rewards loyalty (miners who stay with the pool longer)
- Includes transaction fees in rewards

Disadvantages:

- Variable income depending on pool luck
- Requires consistent mining to maximize returns
- Payment delay until blocks are found and confirmed

Risk Level: Medium

4. Proportional (PROP)

Description: Rewards are distributed proportionally to the number of shares contributed during a round (time between finding blocks).

Advantages:

- Transparent reward calculation
- Lower fees than PPS
- Payments include transaction fees

Disadvantages:

- Encourages pool hopping (switching between pools)
- Variable income depending on round length and luck
- Payment delay until blocks are found and confirmed

Risk Level: Medium

5. Solo Mining Pools

Description: Provides infrastructure for solo mining while allowing miners to use the pool's node.

Advantages:

- Full block reward if successful (minus small pool fee)
- No dilution of rewards with other miners
- Privacy and independence

Disadvantages:

- Extremely irregular payouts
- High variance (may go months without finding a block)
- Only viable for miners with substantial hash power

Risk Level: Very High

6. Merge Mining Pools

Description: Allows miners to mine Bitcoin and compatible alternative cryptocurrencies simultaneously.

Advantages:

- Additional revenue from altcoins without additional computing power
- Diversification of mining rewards
- Potential for higher overall returns

Disadvantages:

- More complex setup and configuration
- Additional overhead in managing multiple coin wallets
- Rewards from altcoins may fluctuate significantly

Risk Level: Medium

Key Decision Factors

1. Hash Power and Expected Returns

- Large-scale operations (>100 TH/s):
 - Can consider any pool type, including PPLNS for maximum returns
 - May even consider solo mining if hash power exceeds 1% of network
- Medium-scale operations (10-100 TH/s):
 - PPLNS or FPPS recommended for balance of risk and reward
 - Avoid solo mining due to high variance
- Small-scale operations (<10 TH/s):
 - PPS or FPPS recommended for consistent income
 - High variance payment systems may result in long periods without payment

2. Pool Size and Network Share

- Large pools (>10% network hash rate):
 - More consistent block discovery
 - Higher risk of centralization concerns
 - Potential target for regulatory actions
- Medium pools (1-10% network hash rate):
 - Good balance between consistency and decentralization
 - Generally stable operations and infrastructure
- Small pools (<1% network hash rate):
 - Support network decentralization
 - Higher variance in block discovery
 - May have less robust infrastructure

3. Pool Fees

Payment Model	Typical Fee Range
PPS	2-4%

FPPS	2-3%
PPLNS	1-2%
PROP	0-1%
Solo	0-2%

4. Geographic Considerations

- Server Location: Choose pools with servers near your mining operation to minimize latency
- Jurisdiction: Consider the regulatory environment of the pool's operating country
- Language Support: Ensure documentation and support are available in your preferred language

5. Security and Reliability

- Uptime History: Research the pool's historical uptime percentage
- Security Measures: Two-factor authentication, SSL connections, cold storage of funds
- Track Record: Length of operation, past security incidents, response to issues
- Payout Thresholds: Lower thresholds reduce risk exposure but increase transaction fees

6. Pool Transparency

- Open Source Software: Indicates transparency and allows community review
- Real-time Statistics: Detailed visibility into pool performance and share accounting
- Communication: Regular updates and clear communication during incidents

Risks to Consider

1. Pool Centralization

- Network Security Risk: Pools approaching 51% of hash rate threaten Bitcoin's security
- Ethical Consideration: Supporting diverse pools helps maintain Bitcoin's decentralization
- Regulatory Target: Larger pools may face increased regulatory scrutiny

2. Pool Operator Risks

- Exit Scams: Operators could potentially disappear with unpaid rewards
- Mismanagement: Poor security practices could lead to pool hacks
- Fee Changes: Sudden changes in fee structures without proper notice

3. Technical Risks

- Stale Shares: High latency connections can increase rejected shares
- Downtime: Pool server outages result in lost mining time
- Software Bugs: Errors in pool software can affect proper reward distribution

4. Financial Risks

- Payment Delays: Some pools have significant delays between mining and payment
- Minimum Payout Thresholds: Small miners may take extensive time to reach minimums
- Wallet Security: Consider the security of the wallet address receiving payouts

Decision Flowchart

- Determine your priority:
 - Consistent income → PPS/FPPS
 - Maximum returns → PPLNS
 - Supporting decentralization → Smaller pools
 - Full independence → Solo mining (if hash power permits)
- Evaluate your risk tolerance:
 - Low risk → Large pool with PPS/FPPS
 - Medium risk → Medium-sized pool with PPLNS
 - High risk → Small pool or solo mining
- Consider technical factors:
 - Server proximity
 - User interface requirements
 - Monitoring tools needed
- Research pool reputation:
 - Community feedback
 - Historical reliability
 - Transparency practices

Top Considerations Summary

1. Payment model matching your needs and risk profile
2. Pool size balancing consistency with decentralization values
3. Geographic proximity for minimizing latency
4. Security features and historical reliability
5. Fee structure and minimum payouts
6. Transparency and communication quality

Recommended Evaluation Process

1. Create a shortlist of 3-5 pools based on payment model preference
2. Compare fee structures and estimated returns
3. Research each pool's security history and user reviews
4. Test each pool for a limited time period (1-2 weeks)
5. Evaluate actual returns and technical performance
6. Make final decision based on actual experience

Remember that pool conditions change over time, so it's advisable to reassess your mining pool choice periodically as your operation grows and market conditions evolve.

Top Bitcoin Mining Pools by Type

Pay-Per-Share (PPS) / Full Pay-Per-Share (FPPS)

1. **Binance Pool** - Primarily offers FPPS, operated by the Binance cryptocurrency exchange
2. **Poolin** - Offers both PPS+ and FPPS options, major pool with comprehensive mining services
3. **BTC.com** - Operates with FPPS model, previously owned by Bitmain
4. **KuCoin Pool** - Uses PPS+ payment system, associated with the KuCoin exchange
5. **Huobi Pool** - Primarily FPPS-based, associated with Huobi exchange
6. **OKEx Pool** - Uses PPS+ payment model, operated by the OKEx cryptocurrency exchange

Pay-Per-Last-N-Shares (PPLNS)

1. **Foundry USA** - Uses PPLNS with some customizations, US-based pool that became dominant in 2021-2024
2. **AntPool** - Offers both PPLNS and PPS options, operated by Bitmain
3. **F2Pool** - Primarily PPLNS-based, one of the longest-running pools
4. **Braiins Pool (Slush Pool)** - Uses a modified PPLNS called "Score-based" system, the first Bitcoin mining pool ever created
5. **EMCD** - Uses PPLNS system, European-based mining pool with global presence
6. **Luxor** - Primarily PPLNS-based, North American pool focusing on transparent mining services
7. **NovaBlock** - Operates with PPLNS model, newer pool that gained significant hashrate
8. **Sigmapool** - Uses PPLNS system, smaller but established pool with loyal miners
9. **Ocean Pool** - Uses PPLNS with Stratum V2 protocol, operated by Braiins (same company as Slush Pool), focuses on decentralization and giving miners more control

Mixed/Multiple Payment Options

1. **ViaBTC** - Offers both PPS+ and PPLNS options, provides mining services for multiple cryptocurrencies
2. **1THash** - Provides multiple payment methods, Chinese mining pool with significant hashrate
3. **BTC.TOP** - Offers various payment methods, significant Chinese mining pool
4. **BitDeer** - Offers several payment options as part of their cloud mining services

Solo/Custom

1. **MARA Pool** - Primarily operates for Marathon's own mining operations with custom reward structures, run by Marathon Digital Holdings
2. **SBI Crypto** - Operates with custom payment models for institutional miners, Japanese financial conglomerate's mining operation

Key Features of Top Bitcoin Mining Pools

Pay-Per-Share (PPS) / Full Pay-Per-Share (FPPS) Pools

1. Binance Pool

- Payment Model: FPPS (Full Pay-Per-Share)
- Operator: Binance cryptocurrency exchange
- Fee Structure: 2.5% fee for Bitcoin mining
- Minimum Payout: 0.001 BTC
- Server Locations: Global (Asia, Europe, North America)
- UI/Dashboard: Advanced, integrated with Binance exchange
- Extra Features:
 - Mining revenue can be directly used for trading
 - Integrated with Binance's comprehensive ecosystem
 - Smart Pool feature automatically switches to most profitable coin
- Security: Industry-leading exchange security protocols
- Launched: April 2020

2. Poolin

- Payment Model: PPS+ and FPPS options
- Operator: Poolin, Inc.
- Fee Structure: 2.5% for PPS+, 2% for FPPS
- Minimum Payout: 0.001 BTC
- Server Locations: Global (China, USA, Germany)

- UI/Dashboard: Comprehensive mining statistics and monitoring
- Extra Features:
 - Multi-currency mining support
 - Mobile app for monitoring
 - Merged mining capabilities
 - Mining wallet with financial services
- Security: Cold wallet storage, 2FA
- Launched: November 2017

3. BTC.com

- Payment Model: FPPS
- Operator: Independent (formerly Bitmain)
- Fee Structure: 2.5-4% depending on the coin
- Minimum Payout: 0.001 BTC
- Server Locations: Global (China, USA, Europe)
- UI/Dashboard: Detailed real-time statistics and analytics
- Extra Features:
 - Multi-coin mining platform
 - Advanced API access
 - Block explorer integration
 - Mobile app for monitoring
- Security: Cold storage, 2FA, IP restrictions
- Launched: 2016

4. KuCoin Pool

- Payment Model: PPS+
- Operator: KuCoin cryptocurrency exchange
- Fee Structure: 2% for Bitcoin mining
- Minimum Payout: 0.0005 BTC
- Server Locations: Global nodes with focus on Asia
- UI/Dashboard: Integrated with KuCoin exchange interface
- Extra Features:
 - Direct trading of mined assets
 - Low barrier to entry for beginners
 - Staking options for mining rewards
 - Regular promotions for miners
- Security: Exchange-grade security protocols
- Launched: 2021

5. Huobi Pool

- Payment Model: FPPS
- Operator: Huobi Global exchange

- Fee Structure: 2-3% depending on coin
- Minimum Payout: 0.001 BTC
- Server Locations: Global with strong presence in Asia
- UI/Dashboard: Integrated with Huobi exchange
- Extra Features:
 - Seamless exchange integration
 - Mining tutorials for beginners
 - Token mining options beyond Bitcoin
 - Regular mining campaigns and incentives
- Security: Exchange security infrastructure
- Launched: 2018

6. OKEx Pool

- Payment Model: PPS+
- Operator: OKEx cryptocurrency exchange
- Fee Structure: 2.5% for Bitcoin
- Minimum Payout: 0.001 BTC
- Server Locations: Global data centers
- UI/Dashboard: Exchange-integrated mining dashboard
- Extra Features:
 - Direct integration with OKEx exchange
 - Multi-currency mining options
 - Financial services for mining profits
 - Mining data analysis tools
- Security: Exchange-level security protocols
- Launched: 2019

Pay-Per-Last-N-Shares (PPLNS) Pools

7. Foundry USA

- Payment Model: PPLNS with customizations
- Operator: Digital Currency Group (DCG)
- Fee Structure: 1-2% (negotiable for large miners)
- Minimum Payout: 0.0005 BTC
- Server Locations: North America focused
- UI/Dashboard: Enterprise-grade monitoring and analytics
- Extra Features:
 - Institutional focus
 - Regulatory compliance for US miners
 - Integration with Foundry Digital Asset Mining solutions
 - Access to equipment financing
- Security: Institutional-grade security

- Launched: 2020

8. AntPool

- Payment Model: PPLNS and PPS options
- Operator: Bitmain Technologies
- Fee Structure: 0.5-2.5% depending on payment method
- Minimum Payout: 0.001 BTC
- Server Locations: Global (China, USA, Europe, Russia)
- UI/Dashboard: Feature-rich statistics and monitoring
- Extra Features:
 - Multi-currency support
 - Mining monitor app
 - ASIC integration (Bitmain hardware)
 - Regular promotions for miners
- Security: Advanced 2FA, cold wallet storage
- Launched: 2014

9. F2Pool

- Payment Model: PPLNS
- Operator: F2Pool team
- Fee Structure: 2.5% for Bitcoin
- Minimum Payout: 0.001 BTC
- Server Locations: Global (China, USA, Europe, Russia)
- UI/Dashboard: Clean interface with comprehensive statistics
- Extra Features:
 - Extensive multi-coin support
 - Personalized hashrate monitoring
 - Mining calculator tools
 - Strong community support
- Security: 2FA, cold storage, IP restrictions
- Launched: 2013

10. Braiins Pool (Slush Pool)

- Payment Model: Score-based system (modified PPLNS)
- Operator: Braiins Systems
- Fee Structure: 2% for Bitcoin
- Minimum Payout: 0.001 BTC
- Server Locations: Global (Europe, USA, Asia)
- UI/Dashboard: Pioneer of detailed mining statistics
- Extra Features:
 - Creator of Stratum V2 protocol
 - Braiins OS+ firmware integration

- Advanced monitoring tools
 - Historical performance tracking
- Security: Industry-leading security practices
- Launched: 2010 (first Bitcoin pool ever)

11. EMCD

- Payment Model: PPLNS
- Operator: EMCD Services Ltd.
- Fee Structure: 1.5% for Bitcoin
- Minimum Payout: 0.001 BTC
- Server Locations: Europe, Russia, Central Asia
- UI/Dashboard: User-friendly interface with detailed statistics
- Extra Features:
 - Multi-language support (focus on Eastern Europe)
 - Daily automatic payments
 - Telegram bot integration
 - Referral program
- Security: 2FA, wallet verification
- Launched: 2018

12. Luxor

- Payment Model: PPLNS
- Operator: Luxor Technology Corporation
- Fee Structure: 0.5-2% (negotiable for large miners)
- Minimum Payout: 0.001 BTC
- Server Locations: North America focused
- UI/Dashboard: Advanced hashrate and earnings visualization
- Extra Features:
 - Switching algorithm for maximum profit
 - Hashrate marketplace
 - Proprietary mining software
 - North American focus with regulatory compliance
- Security: Industry standard security
- Launched: 2017

13. NovaBlock

- Payment Model: PPLNS
- Operator: NovaBlock team
- Fee Structure: 1% for Bitcoin
- Minimum Payout: 0.001 BTC
- Server Locations: Global (North America, Asia)
- UI/Dashboard: Clean, intuitive interface

- Extra Features:
 - Focus on network decentralization
 - Transparent operations
 - Regular payment schedule
 - Community engagement
- Security: Comprehensive security protocols
- Launched: 2019

14. Sigmapool

- Payment Model: PPLNS
- Operator: Sigmapool team
- Fee Structure: 0.8-1%
- Minimum Payout: 0.002 BTC
- Server Locations: Europe, Asia
- UI/Dashboard: Simple but effective interface
- Extra Features:
 - Merged mining capabilities
 - Autonomous payments
 - Smaller pool with personalized service
 - Community-oriented approach
- Security: 2FA, email confirmations
- Launched: 2016

15. Ocean Pool

- Payment Model: PPLNS
- Operator: Braiins (operators of Slush Pool)
- Fee Structure: 1-2%
- Minimum Payout: 0.001 BTC
- Server Locations: Global distribution
- UI/Dashboard: Modern interface with detailed metrics
- Extra Features:
 - Stratum V2 protocol implementation
 - Enhanced decentralization focus
 - Miner control over block templates
 - Censorship-resistant mining
 - Transaction selection transparency
- Security: Advanced security from Braiins team
- Launched: 2022

Mixed/Multiple Payment Options Pools

16. ViaBTC

- Payment Model: PPS+ and PPLNS options
- Operator: ViaBTC Group
- Fee Structure: 2% for PPS+, 1% for PPLNS
- Minimum Payout: 0.001 BTC
- Server Locations: Global (China, USA, Europe)
- UI/Dashboard: Comprehensive statistics and analytics
- Extra Features:
 - Multi-coin mining
 - Mining profit converter
 - CoinEx exchange integration
 - Transaction accelerator service
- Security: Cold wallet storage, 2FA
- Launched: 2016

17. 1THash

- Payment Model: Multiple payment methods
- Operator: 1THash team
- Fee Structure: 1-2.5% depending on payment model
- Minimum Payout: 0.001 BTC
- Server Locations: Primarily Asia-Pacific
- UI/Dashboard: Detailed mining statistics
- Extra Features:
 - Focus on Chinese miners
 - Multiple settlement options
 - Regular payment cycles
 - Mining equipment integration
- Security: Standard security protocols
- Launched: 2018

18. BTC.TOP

- Payment Model: Various payment methods
- Operator: BTC.TOP team
- Fee Structure: 1-3% depending on selected model
- Minimum Payout: 0.001 BTC
- Server Locations: Asia-focused with global nodes
- UI/Dashboard: Comprehensive mining dashboard
- Extra Features:
 - Mining contract options
 - Customizable payout thresholds

- Hardware optimization tools
 - Multi-language support
- Security: 2FA, email notifications
- Launched: 2017

19. BitDeer

- Payment Model: Multiple options with cloud mining
- Operator: BitDeer team
- Fee Structure: Varies by contract (2-4%)
- Minimum Payout: Contract dependent
- Server Locations: Global data centers
- UI/Dashboard: User-friendly cloud mining interface
- Extra Features:
 - Cloud mining contracts
 - No hardware maintenance
 - Flexibility in contract selection
 - Multiple cryptocurrency options
- Security: Cold wallet storage, multi-signature
- Launched: 2018

Solo/Custom Pools

20. MARA Pool

- Payment Model: Custom rewards structure
- Operator: Marathon Digital Holdings
- Fee Structure: Private/Custom
- Minimum Payout: N/A (primarily self-mining)
- Server Locations: North America (USA)
- UI/Dashboard: Proprietary mining management system
- Extra Features:
 - OFAC-compliant mining (historically)
 - Institutional-grade infrastructure
 - Publicly traded company backing
 - Renewable energy focus
- Security: Enterprise-level security
- Launched: 2020

21. SBI Crypto

- Payment Model: Custom/Institutional
- Operator: SBI Holdings (Japanese financial conglomerate)
- Fee Structure: Private/Custom
- Minimum Payout: N/A (institutional focus)

- Server Locations: Asia-Pacific
- UI/Dashboard: Enterprise mining management
- Extra Features:
 - Institutional mining focus
 - Financial services integration
 - Corporate governance structure
 - Strategic mining partnerships
- Security: Financial institution security standards
- Launched: 2017

Key Considerations When Comparing Pools

1. Payment Model Alignment: Choose a pool whose payment model matches your risk tolerance and operational scale
2. Geographic Proximity: Select pools with servers close to your mining operation to minimize latency
3. Fee Structure Impact: Consider how fees affect your profitability under different market conditions
4. Payout Thresholds: Higher thresholds mean less frequent payouts but lower transaction fees
5. Pool Size/Hashrate: Larger pools provide more consistent rewards but contribute less to network decentralization
6. UI Requirements: Consider the importance of monitoring tools and statistical information for your operation
7. Additional Services: Evaluate whether exchange integration, mining software, or other services add value
8. Security Track Record: Research the pool's history regarding downtime, hacks, or payment issues
9. Philosophy Alignment: Some pools prioritize censorship resistance and decentralization over pure profit maximization